

Contents

1	Appendix D: Neural Encoding Efficiency on Time-Series	1
1.1	Objective	1
1.2	Methods (src)	1
1.3	Script and Outputs	1
1.4	Figure	2
1.5	Equation References	2
1.6	Reproducibility	2
1.7	Cross-References	3

1 Appendix D: Neural Encoding Efficiency on Time-Series

1.1 Objective

Comprehensive neural encoding analysis including spike train generation, temporal dynamics, population coding, mutual information, and adaptation mechanisms for olfactory receptor neurons.

1.2 Methods (src)

- `src/case_studies/neural_encoding.py`
 - `generate_spike_trains(stimuli, dt, baseline_rate, max_rate, dynamics)` - Realistic spike generation
 - `analyze_spike_train_statistics(spike_data)` - ISI, CV, Fano factor analysis
 - `temporal_coding_analysis(spike_data, stimulus_times)` - Latency and precision metrics
 - `population_coding_analysis(population_responses, labels)` - PCA, LDA, correlations
 - `mutual_information_analysis(responses, stimuli)` - Information theory metrics
 - `odor_discrimination_analysis(responses, odor_ids, time_windows)` - Discrimination performance
 - `adaptation_dynamics_analysis(spike_data, stimulus_duration)` - Adaptation characterization
 - `information_rate_time_series(responses, dt_s, noise_std)` - Channel capacity estimation
 - `rate_coding_metrics(responses, labels)` - Basic discriminability metrics

1.3 Script and Outputs

- Script: `scripts/generate_neural_encoding_analysis.py`

- Data: output/data/neural_encoding_comprehensive.npz
- Figure: output/figures/neural_encoding_comprehensive_analysis.png

1.4 Figure

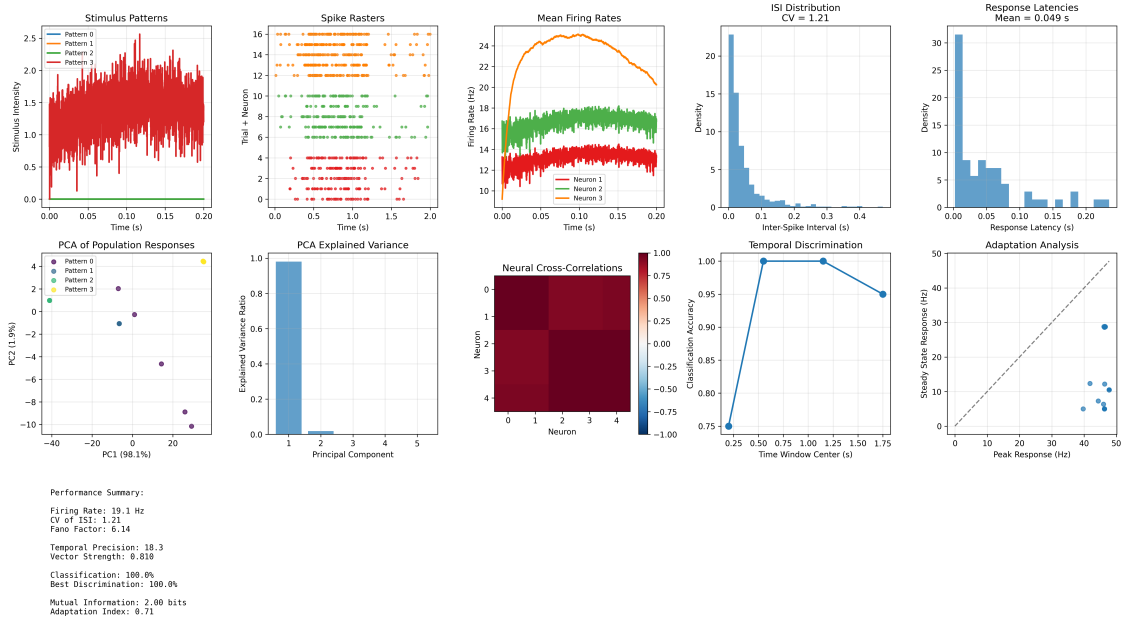


Figure 1: Comprehensive neural encoding analysis: Multi-neuron spike train generation with diverse response dynamics (exponential, adaptive, phasic-tonic), population coding with PCA dimensionality reduction, temporal precision analysis, inter-spike interval statistics, mutual information quantification, and adaptation dynamics characterization. Includes cross-correlation analysis, discrimination performance across time windows, and information-theoretic metrics for olfactory neural processing.

1.5 Equation References

- Information rate: see (??)
- Response time model: see (??)

1.6 Reproducibility

- Run: `python3 scripts/generate_neural_encoding_analysis.py`
- Artifacts saved to output/data/ and output/figures/.
- Deterministic seeds: `src/config.set_random_seed(42)` for surrogate time-series.

1.7 Cross-References

- Methods: ??
- Symbols: ??
- Math appendix: ??